D 32228 **(Pages : 4)** **Name.....**

Reg. No.....

FIRST SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2022

Mathematics

MAT 1C 01—MATHEMATICS

(2017 and 2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A (Objective Type Questions)

Answer **all** *questions* (1 - 12).
Each question carries 1 *mark.*

1. $\lim_{x \to 0} \frac{\tan x}{x} =$

2. If $\lim_{x \to 4} \frac{f(x) - 5}{x - 2} = 1$, then $\lim_{x \to 4} f(x) =$ _____.

3. State intermediate value theorem.

4. Define critical point.

5. What is the physical interpretations of the mean value theorem.

6. State the second derivative test for local extreme values of a function $y = f(x)$.

7. If $y = e^{\sin x} - \log \cos x + 2$, $dy =$

8. If f and g are polynomials with $\deg(f) < \deg(g)$, what is $\lim_{x \to \infty} \frac{f(x)}{g(x)}$?

9. Define asymptote of the graph of a function.

10. State Max-Min Inequality of definite integrals.

Turn over

11. If f is an odd function, $\int_{-a}^{a} f(x)\,dx =$ _____.

12. When we say that a function $y = f(x)$ is concave down in $[a, b]$?

(10 × 1 = 10 marks)

Part B (Short Answer Type)

Answer any **nine** *questions* (13 - 24).
Each question carries 2 *marks.*

13. Using formal definition of limit, show that $\lim_{x \to 2} f(x) = 4$, where $f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2 \end{cases}$.

14. Show that $y = \sin(1/x)$ has no limit as x approaches zero from either side.

15. Find the equation of the tangent line to the curve $y = x + \frac{2}{x}$ at $x = 1$.

16. Show that composite of two continuous function is continuous.

17. Find the equation of the tangent line to $y = \cos x$ at $x = 0$.

18. Find the left and right derivatives of $y = |x|$ at $x = 0$.

19. Find the horizontal/vertical asymptotes of the graph of $f(x) = 2 + \frac{\sin x}{x}$.

20. What is the geometrical significance of Rolle's theorem.

21. What is the physical interpretation of Mean Value Theorem.

22. Suppose that f is continuous on $[a, b]$ and differentiable on (a, b).

If $f' > 0$ at each point of (a, b), then show that f is increases on $[a, b]$.

If $f' < 0$ at each point of (a, b), then show that f is decreases on $[a, b]$.

23. Find the critical points of $y = x^{4/3} - x^{1/3}$.

24. Find the average value of $y = 1/x$ in $[1, 2]$.

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any **six** *questions* (25 - 33).
Each question carries 5 *marks.*

25. Give an example of a function and a point where both left and right hand limits exist but limit does not exist. Also prove your claim.

26. How close to $x_0 = 4$ must we hold the input x to be sure that the output $y = 2x - 1$ lies within 2 units of $y_0 = 7$.

27. If f as a derivative at $x = c$, then show that f is continuous at $x = c$.

28. Find the maxima and minima of $f(x) = \dfrac{(x-1)(x-6)}{x-10}$.

29. Find the points of inflection of $x = \log(y/x)$.

30. Find the intervals in which $y = x^3 + 3x^2 - 9x + 7$ is increasing or decreasing.

31. Find the area of the region between the x-axis and the graph of the function $f(x) = x^3 - x^2 - 2x, \ -1 \le x \le 2$.

32. The household voltage in our home be given by $V = V_{max} \sin(120\pi t)$, where V_{max} is the peak voltage and t is the time in seconds. Find the average voltage over a half-cycle.

33. Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x-axis and the line $y = x - 2$.

(6 × 5 = 30 marks)

Turn over

Part D (Essay Questions)

Answer any **two** *questions* (34 - 36).
Each question carries 10 *marks.*

34. Show that $\lim_{x \to a} (f+g)(x) = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$, provided the two limits on the right side exist.

35. Trace the curve $(x^2 + y^2)x - ay^2 = 0, \ (a > 0)$.

36. State and prove fundamental theorem of Calculus (Part 2).

(2 × 10 = 20 marks)